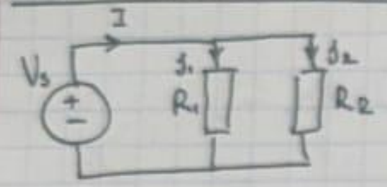
Ele 1802 Quiz 2

Power = ? V^2/R

Q1



$V_s = 171.3$ V

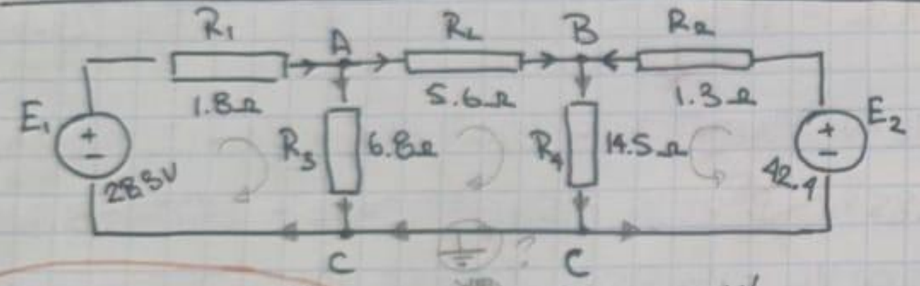
$R_1 = 188.3\,\Omega$ $R_2 = 211.1\,\Omega$

$R_{parallel} \Rightarrow 1/R_L = 1/R_1 + 1/R_2 = 1/188.3 + 1/211 = 0.0100$

$\Rightarrow R_L = 99.5296\,\Omega$

$P = (171.3)^2/99.5296 = 294.8385$ W

Q2



Power @ R_L = ? $P = V^2/R$ $I = V/R$

KCL @A $\frac{V_a - 28.3}{1.8} \overset{-?}{+} \frac{V_a - V_b}{5.6} \overset{-?}{+} \frac{V_a}{6.8} = 0$

$(V_a - 28.3)(5.6 \times 6.8) + (1.8 \times 6.8)(V_a - V_b) + V_a(1.8 \times 5.6) = 0$

$38.08V_a - 1077.664 + 12.24V_a - 12.24V_B + 10.08V_b = 0$

$60.4V_a - 1077.664 - 12.24V_B = 0 \rightarrow V_b = 17.842 + 0.203V_b$

KCL @B $\frac{V_b - V_a}{5.6} + \frac{V_b - 42.4}{1.3} \overset{-?}{+} \frac{V_b}{14.5} = 0$

$(V_b - V_a)(1.3 \times 14.5) + (V_b - 42.4)(5.6 \times 14.5) + V_b(5.6 \times 1.3) = 0$

$18.85V_b - 18.85V_a + 81.2V_b - 3442.88 + 7.28V_b = 0$

$107.33V_b - 3442.88 - 18.85V_a = 0$

$107.33V_b - 3442.88 - 18.85(17.842 + 0.203V_b) = 0$ Sub V_a

$(107.33 - 3.827)V_b - (3442.88 + 336.322) = 0$ LTspice ok's

$103.503V_b = 3779.202 \Rightarrow V_b = 36.513$ $V_b = 25.254$

$P_{R_L} = \frac{(36.513 - 25.254)^2}{5.6} = 22.637$ W

Q3

R_L	I
2.1	10.1
17.0	8.7
Ω	A

V_{AB} = ?

$I = V/(R_1 + R_L)$

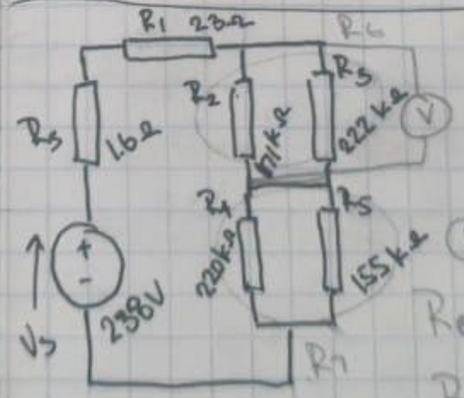
$10.1 = V/(R_1 + 2.1) \rightarrow (10.1)(R_1 + 2.1) = V$

$8.7 = V/(R_1 + 17.0) \rightarrow (8.7)(R_1 + 17.0) = V$

$10.1R_1 + 21.21 = 8.7R_1 + 147.9$

$1.4R_1 = 126.69 \Rightarrow R_1 = 90.493 \Rightarrow V = 935.189$ V

Q4



V_{R2} = ? Voltmeter @ $R_V = 1.18$ MΩ

Measurment error = ?

① Theory.

$R_6 = 1/(1/671 + 1/222) = 96.595$ kΩ

$R_7 = 1/(1/220 + 1/155) = 90.933$ kΩ

$V_{@R6} \Rightarrow \frac{96.595 \times 10^3}{96595 + 90933 + 2341.6} \times 238 = 122.577$ V